Computers and Electronics in Agriculture 227 (2024) 109596

Contents lists available at ScienceDirect

Computers and Electronics in Agriculture

journal homepage: www.elsevier.com/locate/compag

Path planning of greenhouse electric crawler tractor based on the improved A* and DWA algorithms

Huiping Guo [a,b,*], Yi Li [a,b], Hao Wang [a,b], Chensi Wang [a,b], Jiao Zhang [c], Tingwei Wang [a,b], Linrui Rong [a,b], Haoyu Wang [a,b], Zihao Wang [a,b], Yaobin Huo [d], Shaomeng Guo [d], Fuzeng Yang [a,b]

[a] College of Mechanical and Electronic Engineering, Northwest A & F University, Yangling, Shaanxi 712100, China
[b] Northern Agricultural Equipment Scientific Observation and Experimental Station, Ministry of Agriculture and Rural Affairs, Yangling, Shaanxi 712100, China
[c] BYD Automobile Company Limited, Xi'an, Shaanxi 710118, China
[d] Xi 'an Dongyang Machinery Company Limited, Xi'an, Shaanxi 710300, China

ARTICLE INFO

Keywords:
Electric tractor
Path planning
Improved A* algorithm
DWA algorithm

ABSTRACT

To improve the intelligence level and the navigation efficiency of electric crawler tractors in facility greenhouses, this paper proposes a path planning algorithm based on the fusion of the improved A* algorithm and the DWA algorithm. The weight coefficients are integrated into the heuristic function of the A* algorithm, the key point selection strategy is improved, and the second-order Bessel curves are used to smooth the path trajectories. Besides, the DWA algorithm is integrated, and the key point of global paths planned by the improved A* algorithm is taken as an interpolation point. This addresses the issue that the traditional A* algorithm needs to search many nodes and has a low computational efficiency, with many path turning points and unsmooth paths. The results of simulation experiments proved that the improved A* algorithm is less time-consuming and obtains more smoother path than the Dijkstra, RRT, and traditional A* algorithms. Meanwhile, tests in a facility greenhouse show that the electric crawler tractor can realize autonomous navigation and obstacle avoidance, with a maximum lateral deviation of 11.20 cm and a maximum heading deviation of 13°, which can meet the requirements of actual operation in facility greenhouses.

1. Introduction

Facility agriculture is labor-intensive. The environment of China's facility agriculture is small, closed, hot, and humid, and the exhaust produced during operation poses risks to both human health and crop growth. At present, China's agricultural machinery facilities have a low level of intelligence and support and lack specialized miniature operational machinery. Also, few intelligent operating machinery can adapt to cultivation in greenhouses (Zhang and Liu, 2015). With the increasing scale of facility agriculture in China, the demand for intelligent facility agricultural machinery is also growing. The research of automatic navigation technology can greatly improve the intelligence level of facility machinery (Duan et al., 2022; Mao et al., 2022; Tan et al., 2020). Path planning is one of the key technologies of electric tractor navigation in facility environments. According to the degrees of environmental information perception, path planning can be divided into global path planning based on known environmental information and local path planning based on real-time sensor data (Cabreira et al., 2019; Chu et al., 2012; González et al., 2016; Liu et al., 2023; Zhao et al., 2023).

Common global path planning algorithms include Dijkstra algorithm, RRT algorithm (Rapidly-exploring Random Trees), and A* algorithm. The Dijkstra algorithm is based on the greedy strategy, and searches outwards layer by layer until the target is reached. The algorithm has a large amount of calculation, complex calculation, and slow search speed (Nasiboglu, 2022). The RRT algorithm randomly samples nodes in the state space through the incremental growth method, and can always obtain a safe collision-free path from the current position to the target position. The algorithm has a wide search range, but it has the disadvantages of low search efficiency, easy curling of the planned path, and non-optimal path (Cao et al., 2019; Meng et al., 2022; Wang et al., 2022; Yi et al., 2022). A* algorithm is a global path planning method based on heuristic function. By evaluating the value of each node, the

* Corresponding author at: College of Mechanical and Electronic Engineering, Northwest A & F University, Yangling, Shaanxi 712100, China.
E-mail address: imguohp@nwafu.edu.cn (H. Guo).

https://doi.org/10.1016/j.compag.2024.109596
Received 1 October 2023; Received in revised form 4 September 2024; Accepted 25 October 2024
Available online 29 October 2024

next best node is obtained until the final target point is reached. The algorithm can find the optimal path, but there are problems such as uneven path (Cao et al., 2021). The common local path planning algorithms include DWA (Dynamic Window Approach) algorithm and artificial potential field method. The DWA algorithm can achieve real-time dynamic obstacle avoidance, but it is easy to fall into local optimum (Fox et al., 1997); the artificial potential field method is conducive to real-time control and is widely used in robot obstacle avoidance, but it is not suitable for robot planning with high degree of freedom (Zhou and He, 2021).

As a typical node search algorithm, A* algorithm has strong practicability in different occasions. The A* algorithm can effectively plan the shortest working path for the known working environment, but the algorithm has many search nodes, slow search speed, many path inflection points, and the path is not smooth, which is not conducive to robot movement (Huang et al., 2017; Li et al., 2021; Xin et al., 2014). In view of the shortcomings of the traditional A* algorithm, scholars have proposed a variety of improved methods. Jeon et al. (2021) used the A* algorithm for path planning for paddy tractors, so that the tractor can accurately reach the destination through the import and export path in positioning and heading. Two path smoothing methods, line-of-sight path smoothing (LOPS) and collinear node path smoothing (CNS), reduce the redundant nodes derived from the A* algorithm. Liu et al. (2022) proposed an optimized A* algorithm for home robots, which sets the evaluation function of the in-situ obstacle node to infinity, so that the robot no longer returns to the starting point when re-routing. The results show that the path planning time of the optimized A* algorithm is 60 % shorter than that of the traditional A* algorithm, which improves the search efficiency, but does not solve the problem of unsmooth path. Wang et al. (2023) proposed an improved A* algorithm for vehicles. Firstly, a collision cost heuristic function based on obstacle position is designed. Secondly, a steering cost heuristic function based on heading angle difference control is established. The results show that the improved A* algorithm can output a smoother path, avoid vehicle contour collision, and improve vehicle safety. Ou et al. (2022) proposed a path planning method based on improved A* algorithm. The improved algorithm introduces path smoothing strategy and security protection mechanism to eliminate redundant points and minimum corner points, and introduces guidance cost model and adaptive cost function. The results show that the algorithm reduces the average search expansion nodes by 11 % and the average path search time by 13 % in the path planning process, which improves the problems of too many inflection points and slow search speed in the path planning of the traditional A* algorithm. Xiong et al. (2021) proposed a path processing method and path tracking method of A* algorithm for intelligent vehicles. Firstly, a safe redundant space configuration method considering vehicle contour is given. Then, the path is generated based on A* algorithm and smoothed by Bessel curve, and the speed is planned according to the path curvature. The results show that the algorithm makes the path planning smooth. Tang et al. (2021) proposed a geometry-based A* algorithm for automated guided vehicles. The algorithm deletes nodes that do not meet the requirements, reduces path nodes, and replaces the broken line at the turning point with a cubic B-spline curve to smooth the path. The results show that the number of nodes is reduced by 10 % to 40 %, and the number of turns is reduced by 25 %. Zhang et al. (2017) proposed a directional A* algorithm for greenhouse robots, using the ' field of view ' smoothing principle, the ' arc-line-arc ' turning strategy and the binary heap acceleration algorithm. The results show that the directional A* algorithm improves the path smoothness, and the path planning speed is increased by an average of 4–7 times. Cheng et al. (2017) proposed an improved A* algorithm for mobile robots, proposed a key point selection strategy, eliminated redundant turning points, and improved the smoothness of the path. Zhao et al. (2018) combined the jump point search strategy to optimize the A* algorithm, screening representative jump points to expand, reducing memory usage and improving computational efficiency, but the planned path inflection points are too many.

The above improvement method is only an improvement of the global path planning algorithm A* algorithm, which does not take into account the unknown obstacles and does not have the ability of local dynamic obstacle avoidance. The DWA algorithm can plan local obstacle avoidance in real time, which is suitable for robot path planning in dynamic environment, but it can not meet the global path optimization. Therefore, it is difficult to perfectly solve the problem of navigation and obstacle avoidance in path planning by using only a single algorithm. Different algorithms should be used for fusion to achieve the complementary advantages and disadvantages of the algorithms, so as to achieve the optimal path planning. To this end, some researchers have integrated A* algorithm and DWA algorithm, which not only considers dynamic obstacles, but also ensures the completeness of path planning.

Wu and Guo (2020)transformed the 8 search directions of the traditional A* algorithm into 5, which improved the efficiency of path search, and set the safety threshold and curve smoothing function to realize the algorithm fusion. The planned path is smoother, but it does not consider the situation that the remaining 5 search directions have obstacles and judge them, and may fall into a dead zone. Yang and Liu (2021) expanded the 8 search neighborhood to 32 neighborhood, the path is smoother, and is integrated with the DWA algorithm, but the calculation amount is large, and the search efficiency needs to be improved. Lao et al. (2021) proposed a key point selection strategy to improve the A* algorithm, and constructed a dynamic evaluation function. The improved algorithm and the DWA algorithm were fused to ensure the smoothness and feasibility of the path. Although the algorithm improved the dynamic window evaluation function and increased the distance function from the end of the robot to the end point, it did not distinguish the type of obstacle, and the ultrasonic wave would have noise points and large errors. Liu et al. (2021) integrated the improved A* algorithm with the DWA algorithm, and designed a dynamic window evaluation function. However, the improved A* algorithm is difficult to deal with the trap map, and there is no specific strategy to deal with dynamic obstacles. Jin and Wang (2023) introduced the ratio of the distance between the current node and the starting point and the target point into the heuristic function to optimize the heuristic function to improve the search efficiency. At the same time, the neighborhood search principle was simplified to 5-domain search to reduce the traversal nodes, and the redundant node improvement method was used to reduce the degree of path transition. The improved A* algorithm was combined with the DWA algorithm to achieve less access nodes, higher search efficiency, smoother path, and obstacle avoidance. Yin et al. (2023) proposed a fusion algorithm combining motion-constrained A* algorithm and DWA algorithm to optimize nodes, improve heuristic function and quadratic redundancy, so as to improve the efficiency of path planning. Compared with the traditional fusion algorithm, the path length is shortened by 3.6 % and the time is shortened by 6.7 %. Wu et al. (2022) combined the improved A* algorithm with the improved DWA algorithm, improved the A* algorithm by reducing the turning angle and the number of turns, and improved the DWA algorithm by introducing a new evaluation function, so as to solve the dynamic obstacle avoidance problem and improve the path smoothness. Xu and Yuan (2022) proposed a robot path planning algorithm combining A* algorithm and adaptive DWA algorithm to solve the problem that the traditional A* algorithm cannot perfectly realize the global path planning algorithm and avoid its own unknown obstacles. First, the safety distance judgment is introduced into the A* algorithm. At the same time, the dynamic tangential method is used to smooth the A* algorithm, and then it is integrated with the adaptive DWA algorithm to realize the global optimal path planning and obstacle avoidance.

The research primarily examines the application of technologies such as household cleaning robots, mobile cars, port guide vehicles, and small autonomous navigation vehicles. By refining the A* algorithm, this study has managed to reduce search nodes, enhance search efficiency, and facilitate smoother processing. This advancement allows the devices

Table 1
Effect of $h(n)$ values on A* algorithm.

$h(n)$ takes on values	Implications
$h(n) = 0$	A* algorithm degenerates into Dijkstra algorithm, which can find the shortest path, but it takes a long time.
$h(n) <$ real cost	Between Dijkstra and A* algorithm, the shortest path can be found. Theoretically, the smaller $h(n)$ is, the more redundant nodes are.
$h(n) =$ real cost	The algorithm has the best performance, no redundant nodes, the fastest running speed, and the shortest path can be found.
$h(n) >$ real cost	Between the A* algorithm and the best-first search algorithm, theoretically, the larger the $h(n)$, the faster the running speed, but the path is not the shortest.
$h(n) \gg$ real cost	The A* algorithm evolves into the best-first search algorithm (BFS, Best-First Search), which has a fast operation speed and a non-shortest path.

to determine globally optimal routes in a single operational mode or through a combination with the DWA algorithm. This integration not only addresses the limitations of the A* algorithm in dynamic obstacle avoidance but also compensates for the DWA algorithm's inability to optimize global paths effectively. The operational environment considered in the above research is characterized by well-maintained, smooth road surfaces where the revised methodologies have notably improved the navigation accuracy of the system and yielded positive research outcomes. However, this study also encompasses the use of an electric crawler tractor in greenhouses, which presents a narrower, longer workspace with a clay road surface—conditions that are significantly more complex and variable than the flat surfaces previously mentioned. Moreover, as a piece of power machinery for greenhouse cultivation, the electric crawler tractor is required to perform diverse functions such as rotary tilling, sowing, harvesting, and transportation. Given the larger size and weight of the vehicle, the currently employed path planning algorithms are inadequate for both the electric crawler tractor and the challenging working environment it operates in.

To address the limitations of the traditional A* algorithm, such as the generation of redundant nodes, low operational efficiency, and uneven paths, as well as the DWA algorithm's susceptibility to local optima, this study proposes a path planning method for greenhouse electric crawler tractors. This method integrates an enhanced A* algorithm with the DWA algorithm, tailored to suit the specific environmental conditions of greenhouse operations, the physical parameters of the electric tractor, and operational requirements. The heuristic function of the traditional A* algorithm has been improved by incorporating a weight coefficient. Additionally, a key point selection strategy along with Bézier curves are employed to optimize the path. This approach minimizes the number of turning points, thereby smoothing the path and enhancing the A* algorithm's performance. Concurrently, the DWA algorithm has been refined by improving the evaluation function. By integrating the enhanced A* algorithm with the modified DWA algorithm, the accuracy of the navigation system for the electric crawler tractor in the greenhouse operating environment is significantly improved, promoting more efficient and accurate path planning in complex and dynamic settings.

2. Improved global path planning of the A* algorithm

2.1. Traditional A* algorithm

A* algorithm is the most effective direct search method for obtaining the shortest path in a static road network. It combines the advantages of the Dijkstra algorithm and the BFS algorithm. While performing a heuristic search to improve search efficiency, it can find an optimal path based on the valuation function (Cao R, 2021). The A* algorithm starts from the initial node and searches the surrounding nodes; the value of each node around is calculated by the evaluation function; the minimum cost node is selected as the new starting search point to continue to search the next node. The above process is repeated until the target point is reached, and then the target point is traced back to the starting point to form a global planning trajectory. In the search process, since each node on the path has the minimum cost, the path cost is the smallest. The cost function of the A* algorithm is:

$$f(n) = g(n) + h(n) \tag{1}$$

where $f(n)$ represents the estimated cost from the initial state to the target state, $g(n)$ represents the cost from the initial state to the current state (already determined), and $h(n)$ represents the estimated cost from the current state to the target state (prediction).

The A* algorithm searches in the direction of approaching the target, and each node in the search direction is investigated in the search process. When a node is reached, its surrounding nodes will be added to the OpenList. The algorithm will select the node with the least estimated value in OpenList as the next extended node and add it to ClosedList at the same time. This process is repeated until the target node is added to OpenList, and the pathfinding is successful.

2.2. A* algorithm improvement strategy

The traditional A* algorithm, while robust, often traverses numerous redundant nodes during the path search, leading to inefficiencies. Furthermore, the paths generated tend to have multiple turning points, resulting in a lack of smoothness. To address these issues, this paper presents enhancements to both the heuristic function and the method for selecting key points within the context of the traditional A* algorithm. These improvements are designed to accelerate the search process, reduce the number of turns in the path, and create smoother trajectories, thus enhancing overall algorithmic performance in pathfinding tasks.

The overall process of the improved A* algorithm in this paper is shown as follows:

2.2.1. Improved heuristic function

When searching the path, the traditional A* algorithm will traverse many redundant nodes so the operation efficiency is low. In the A* algorithm, the selection of the heuristic function $h(n)$ controls the search and expansion of nodes and then the speed and accuracy of the algorithm. The closer the distance estimation value in the algorithm is to the actual value, the faster the search speed will be. When $h(n) = 0$, the A* algorithm will be transformed into the Dijkstra algorithm, which will find the shortest path, but it needs to calculate a large number of nodes, leading to a low efficiency. If $h(n)$ is smaller than the actual cost of moving from n to the target, the A* algorithm will find the shortest path. The smaller the value of $h(n)$ is, the more nodes A* extends, and the slower it runs. If the value of $h(n)$ is equal to the cost of moving from n to the target point, the A* algorithm only needs to find the optimal path without extending any other nodes, and the running speed will be very fast. If the value $h(n)$ is larger than the actual cost of moving from n to the target point, the A* algorithm may not find the shortest path, but it runs faster; if $g(n) = 0$, the A* algorithm will be transformed into the BFS algorithm, and the calculation speed is fast, but the optimal solution may not be found. Therefore, to enhance the accuracy and flexibility of the A* algorithm, it is necessary to improve the heuristic function. In this study, the heuristic function is improved by setting the weight coefficient to reduce the time of path search. The evaluation function of the improved A* algorithm is:

$$f(n) = g(n) + \omega(n)h(n) \tag{2}$$

where $\omega(n)$ represents the weight coefficient of the estimation cost $h(n)$; $h(n)$ selects euclidean distance for calculation.

Considering the actual operation of the tractor, it is crucial to move quickly to the area where the target point is located at the beginning of the search. At the end of the search, it is crucial to obtain the best path to the target point. Thus, when the current point and the target point are far away, the estimated value of the heuristic function is small, which

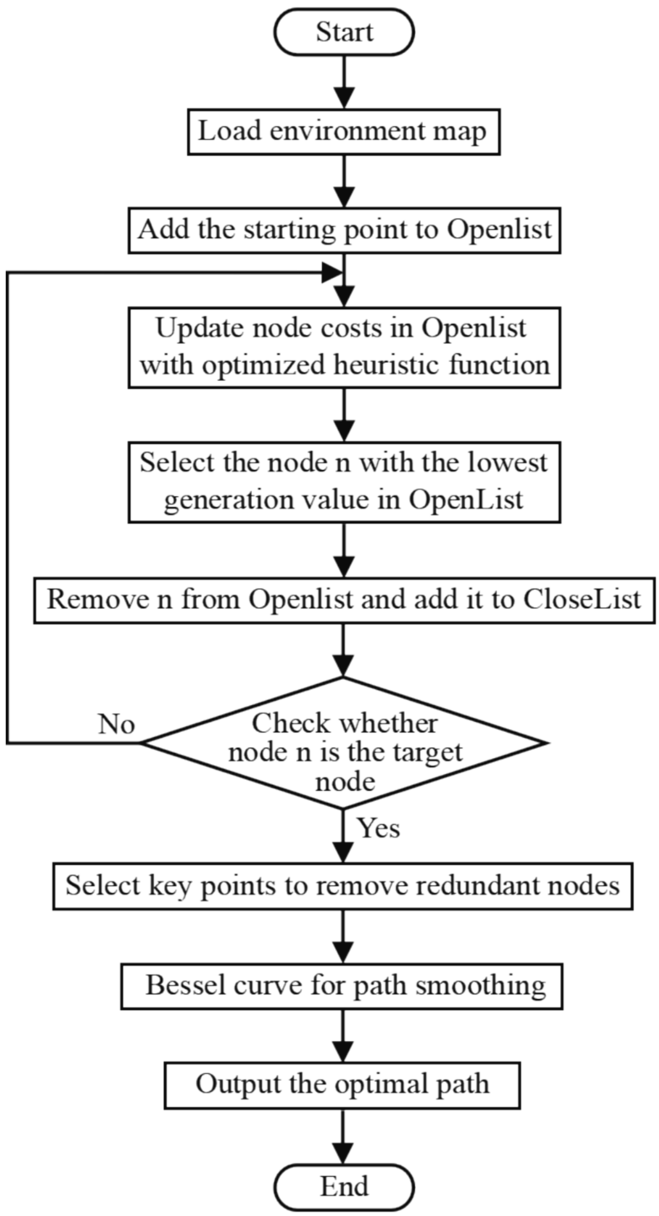


Fig. 1. A* Algorithm improvement flow chart.

may lead to too many search nodes and low efficiency. To improve the computational efficiency, the weight $\omega(n)$ of the estimated value should be increased. When the current point is gradually approaching the target point, the estimated value is gradually approaching the actual one. To prevent the estimated value from being too large to search the optimal path, the estimated value weight $\omega(n)$ should be reduced, thereby reducing the importance of the heuristic function and increasing the importance of the real cost of the path. The value of $\omega(n)$ needs to be adjusted many times according to the size and complexity of the set map, and multi-segment weighting can also be set according to the actual situation (see Table 1).

In order to reduce search time and improve operational efficiency, the improved heuristic function is as follows:

$$f(n) = g(n) + (1 + \frac{d}{D})h(n) \quad (3)$$

Where: d is the distance from the current point to the target point; D is the distance from the start point to the target point.

When approaching the start point, the heuristic function $h(n)$ is less than the actual cost and there are too many redundant nodes. After improvement, the coefficient $(1 + d/D)$ of $h(n)$ is close to 2, the redundant nodes are reduced, the algorithm reduces the search scope and improves the search efficiency. When approaching the target point, the heuristic function $h(n)$ approaches the actual cost, and the coefficient $(1 + d/D)$ of $h(n)$ approaches 1, the algorithm has better performance and fast running speed.

2.2.2. Key point selection strategy

In the actual navigation process, under a single command, the longer the movement distance of the tractor, the better the mobility fluency, the smaller the error, and the more conducive to the accurate completion of the work. The path planned by the traditional A* algorithm consists of continuous grid center points, and there are a large number of redundant collinear nodes and inflection points in the trajectory. When navigated according to the planned path, the tractor needs to adjust the steering angle at each unnecessary turning point. As a result, the path has many turning points, and the path is not smooth. Aiming at these problems, a key point selection strategy is designed in this paper to eliminate redundant nodes and turning points while retaining only key turning points, thereby reducing turning points and improving path smoothness.

The principle of the key point selection strategy is shown in Fig. 2. In the figure, the S point is the starting point, the T point is the target point, the black part represents the obstacle, the white part represents the passable blank area, the black solid line represents the unoptimized path, and the red solid line represents the path optimized by the improved A* algorithm. The path planned by the traditional A* algorithm consists of the connecting lines between the grid center points. The path before optimization is (S, 1, 2, …, 6, T), with a large number of redundant nodes.

The key point selection strategy involves the following steps:

(1) Traversing all the nodes in the path from the starting point S to the target point T in turn. If the current node is on the same line as the two nodes before and after, the current node is regarded as a redundant node and removed. Following this rule, the redundant nodes in the middle of each path are removed, and only the start point, turn point and end point are retained. In Fig. 1, after the intermediate redundant nodes are removed, there are five nodes left: S, 1, 3, 5, and T.

(2) Starting from the next turn point at the start point S, go through each turn point in turn, connecting the point before the turn point and the point after the turn point. If there is obstacles in the middle of the connection, the turn point is retained; If there are no obstacles in the middle, the turn point is deleted, until the turn point before the target point T is traversed. According to this rule, the redundant turn points in the middle of the path are deleted, and only the start point, critical turn point, and target point are retained. In Fig. 1, four nodes S, 3, 5, and T are left after the redundant turn points in the middle are deleted.

(3) Extract the remaining key points, output the optimized path, and obtain the final path composed of nodes S, 3, 5, and T.

After selecting the key points of the improved A* algorithm, the redundant collinear nodes and redundant turn points in Fig. 1. (a) are removed, and the path shown in Fig. 1. (b) is obtained. Since removing redundant nodes will change the original path node set, it is necessary to continuously update the path node set after removing redundant nodes and traverse all path nodes in turn when repeating the above steps. The redundant nodes are removed by the key point selection strategy, and the necessary path nodes are retained. In this approach, the global navigation path containing only the start point, the key point and the target point is optimized, which effectively reduces the length and turn point of the path.

2.2.3. Path smoothing

Since the A* algorithm employs the grid method to represent the environmental map of the facility greenhouse, in the actual operation of the tractor, the curvature at the inflection point of the planned optimal

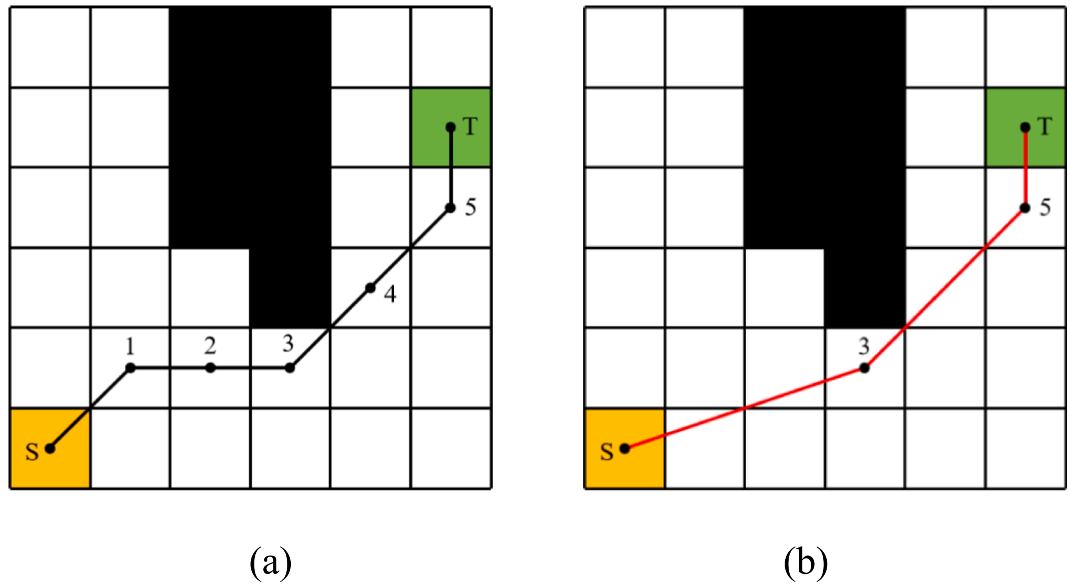


Fig. 2. The schematic diagram of the key point selection strategy: (a) Path before optimization, (b) Optimized path after critical point selection.

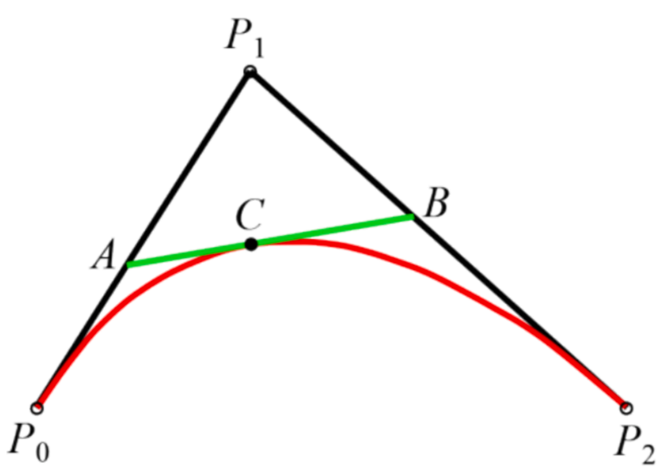


Fig. 3. The schematic diagram of the second-order Bessel curve.

path may be too large, and the speed will change abruptly. In this case, it is easy to break away from the planned path, which will eventually lead to the failure of the tractor navigation task. To make the tractor move smoothly and reduce unnecessary energy consumption at the inflection point of the path, the path trajectory needs to be smoothed after the path planning is completed.

The Bessel curve is a method of displaying a smooth curve in three-dimensional space. When there are $n + 1$ control points, the corresponding Bessel curve of order n is expressed as:

$$P(t) = \sum_{i=1}^{n} P_i B_{i,n}(t), t \in [0,1] \tag{4}$$

$$B_{i,n}(t) = C_n^i t^i (1-t)^{n-i} = \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i}, i = 0, 1, \dots, n \tag{5}$$

where t is normalized time variable; $P(t)$ is the expression of n-order Bessel curve; $B_{i,n}(t)$ is the the basis functions of n-th Bernstein polynomials;

Due to the poor numerical stability of the high-order Bessel curve, to ensure that the tractor can steer smoothly at the inflection point, this study utilizes the second-order Bessel curve to smooth the nodes in the ClosedList of the A* algorithm in turn and then outputs a smooth path.

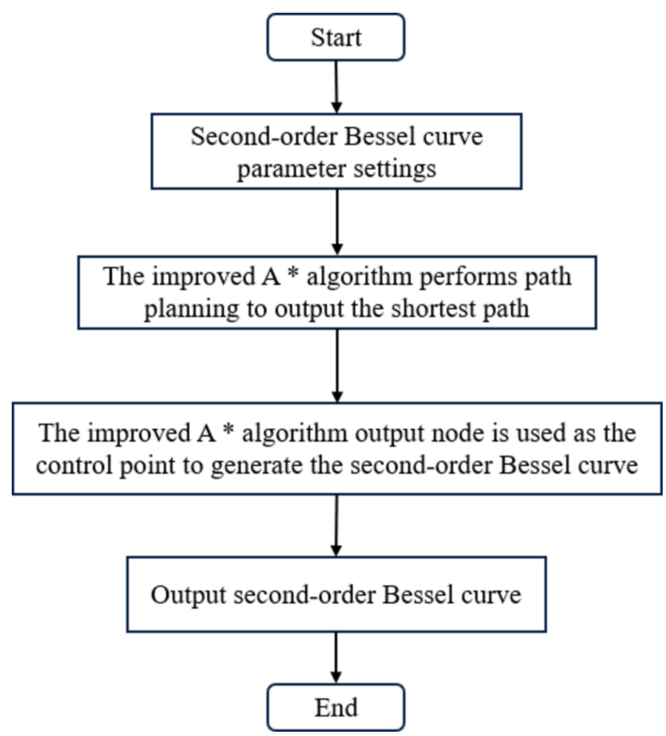


Fig. 4. The flowchart of second-order Bessel curve smoothing paths.

The second-order Bessel curve is shown in Fig. 3:

In the Fig. 3, point A is on line segment P_0P_1, point B is on line segment P_1P_2, point C is on line segment AB, and $t = \frac{P_0A}{AP_1} = \frac{P_1B}{BP_2} = \frac{AC}{CB}$, then the expression of the second-order Bessel curve is:

$$B(t) = (1-t)^2 P_0 + 2t(1-t)P_1 + t^2 P_2, t \in [0,1] \tag{6}$$

where P_0, P_1, and P_2 represent three continuous path points.

The black broken line in Fig. 2 represents the inflection point of the path, and the red curve represents the smoothed path. After the smoothing of the Bessel curve, the original turning point is replaced by a smooth arc so that the path slope at the turning point changes from mutation to gradual variation. It can be seen that when the optimal path

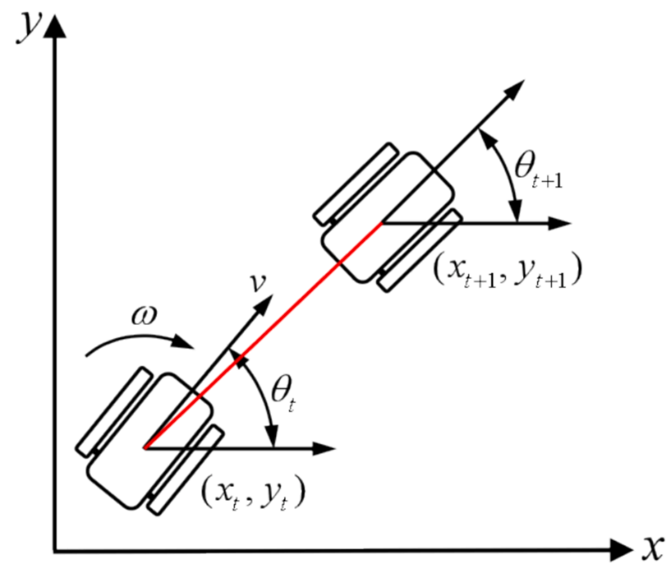


Fig. 5. The kinematics model of the electric crawler differential tractor.

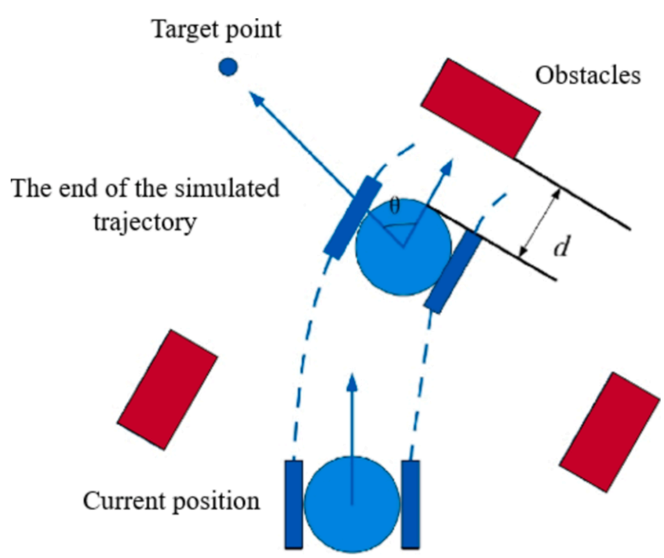


Fig. 6. Tractor azimuth.

is obtained by running the navigation algorithm, the second-order Bessel curve is fitted to each turning point, and eventually multiple Bessel curves are spliced together to obtain a path with significantly reduced curvature and greatly improved path smoothness, which is conducive to reducing the movement loss of the tractor. The flowchart of the second-order Bessel curve smoothing paths is shown in Fig. 4.

3. Local path planning based on improved DWA algorithm

The tractor can navigate well with the global map loaded with complete environmental information. However, in practice, unknown obstacles may appear on the original navigation path of the tractor. Therefore, based on global path planning, this paper utilizes the DWA algorithm to detect the local environment information through the lidar to avoid obstacles in real-time. The DWA algorithm is a local obstacle avoidance algorithm based on kinematics and dynamics theory, and it can convert position control into speed control. It mainly samples multiple sets of speeds in the speed space (linear speed, angular speed) and simulates these speed groups. In the trajectory of the next time

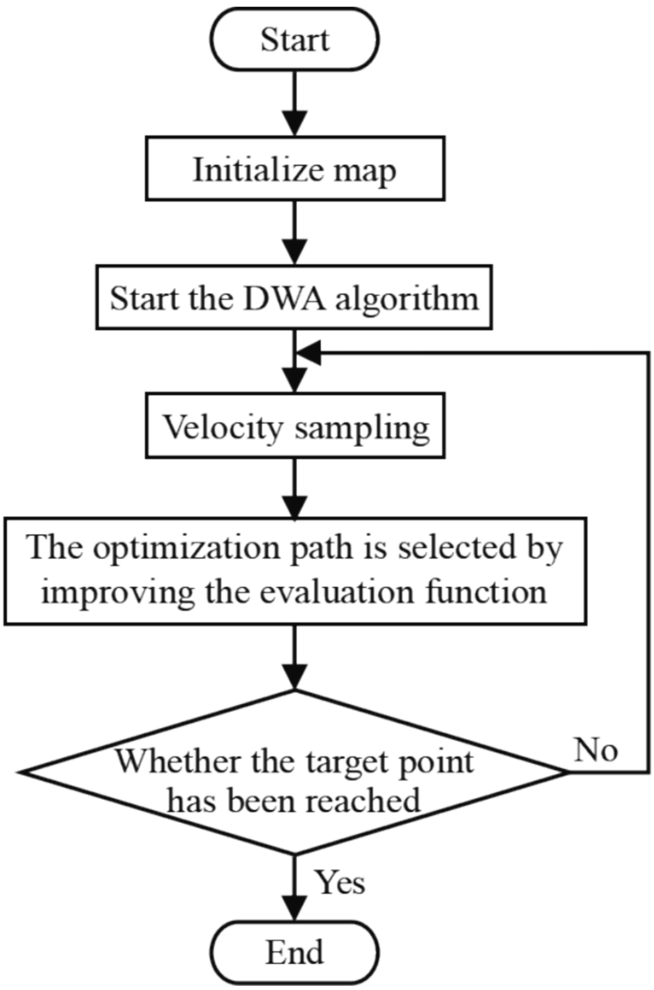


Fig. 7. Flowchart of improved DWA algorithm.

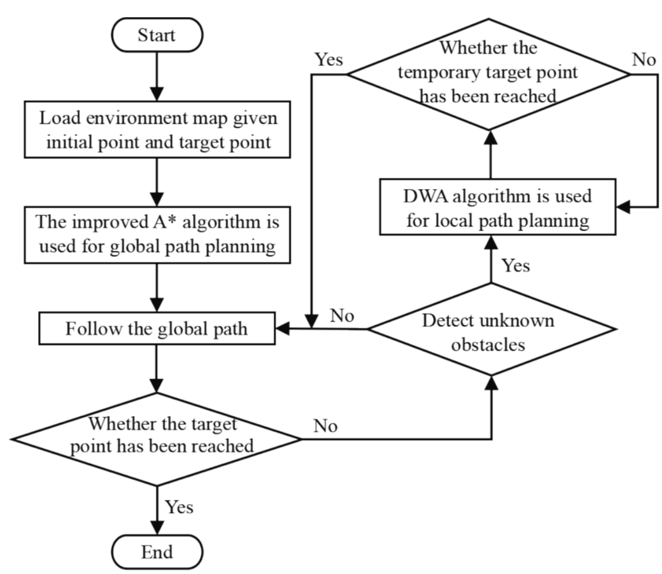


Fig. 8. The flowchart of the fusion algorithm.

interval, the trajectory of collision with obstacles will be eliminated. Finally, an evaluation function is set to score the simulated trajectory, and the speed group corresponding to the optimal simulated trajectory is selected to drive the tractor to complete the corresponding motion.

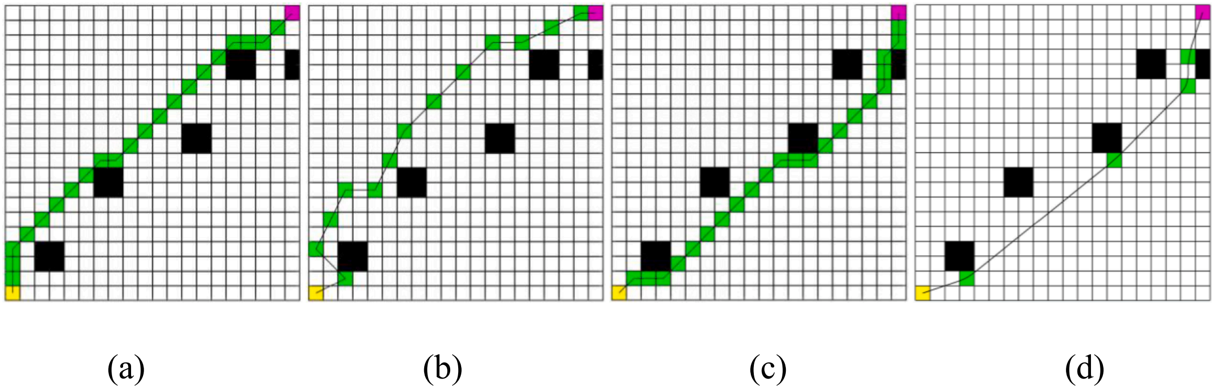


Fig. 9. The path planning simulation results of different algorithms in simple environment: (a) Dijkstra algorithm, (b) RRT algorithm, (c) A* algorithm, (d) Improved A* algorithm.

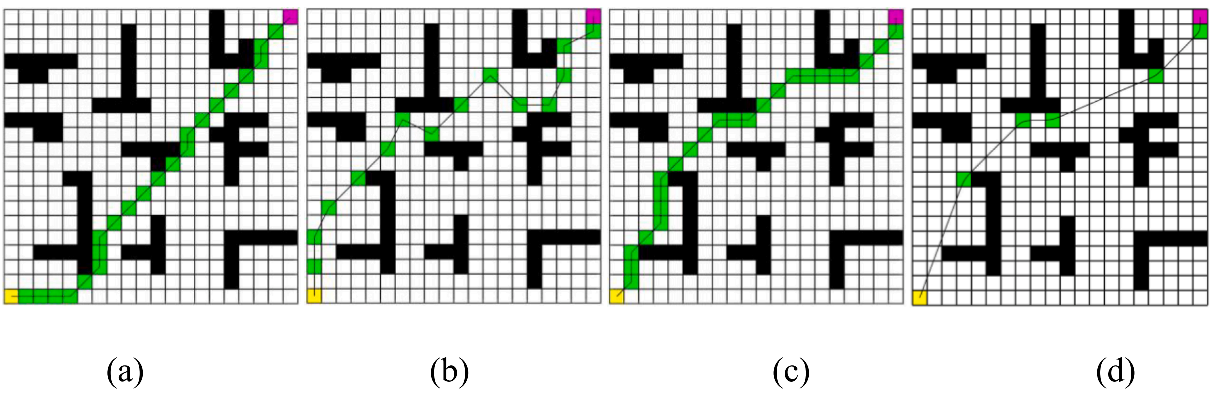


Fig. 10. The path planning simulation results of different algorithms in general environment: (a) Dijkstra algorithm, (b) RRT algorithm, (c) A* algorithm, (d) Improved A* algorithm.

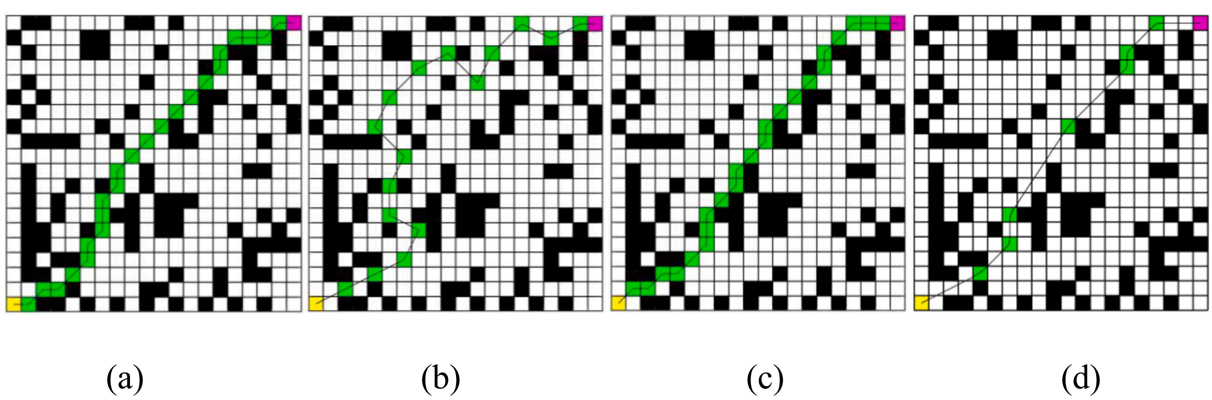


Fig. 11. The path planning simulation results of different algorithms in complex environment: (a) Dijkstra algorithm, (b) RRT algorithm, (c) A* algorithm, (d) Improved A* algorithm.

3.1. Kinematics model

In this paper, the electric crawler tractor is selected as the test platform for experimental validation. Since the DWA algorithm has to transform the position control into speed control, it needs to analyze the tractor motion model, and the kinematics model of the differential motion of the electric crawler tractor is shown in Fig. 5.

The DWA algorithm performs trajectory simulation by modeling the tractor motion to find the best path among several simulated trajectories. Let v and ω be the linear and angular velocities of the tractor at moment t in the world coordinate system, respectively. The variation range of the translational velocity v depends on the nearest distance of the obstacle and the maximum deceleration, and the variation range of the rotational velocity ω depends on the nearest distance of the obstacle and the maximum angular deceleration. In the time interval Δt, assuming the tractor speed is constant, the tractor displacement increment expression is:

$$\begin{cases} \Delta x = v\Delta t\cos\theta_t \\ \Delta y = v\Delta t\sin\theta_t \\ \Delta\theta = \omega\Delta t \end{cases} \tag{7}$$

The bit position at the moment $t+1$ can be expressed as:

Table 2
Comparison of the running time, turn points, and path length of different path planning algorithms in common scenarios.

Parameter	Environment	Dijkstra algorithm	RRT algorithm	Traditional A* algorithm	Improved A* algorithm	RR_1/%	RR_2/%	RR_3/%
Runtime/s	simple environment	0.066356	0.072163	0.058382	0.054215	18.30	24.87	7.14
	general environment	0.064074	0.073817	0.056352	0.052081	18.72	29.44	7.58
	complex environment	0.064821	0.082518	0.057617	0.052863	18.45	35.94	8.25
	average	/	/	/	/	18.49	30.08	7.66
Number of turn points	simple environment	5	8	7	4	20.00	50.00	42.86
	general environment	7	11	9	5	28.57	54.55	44.44
	complex environment	13	14	13	7	46.15	50.00	46.15
	average	/	/	/	/	31.57	51.52	44.48
Path length/cm	simple environment	28.6274	31.9662	29.2132	28.2019	1.49	11.78	3.46
	general environment	29.2132	34.3225	30.3848	29.0593	0.53	15.33	4.36
	complex environment	29.7990	39.4383	29.7990	28.9969	2.69	26.48	2.69
	average	/	/	/	/	1.57	17.86	3.50

Note: RR_1 is the reduction rate of improved A* compared with Dijkstra algorithm, RR_2 is the reduction rate of improved A* compared with RRT algorithm, and RR_3 is the reduction rate of improved A* compared with traditional A* algorithm.

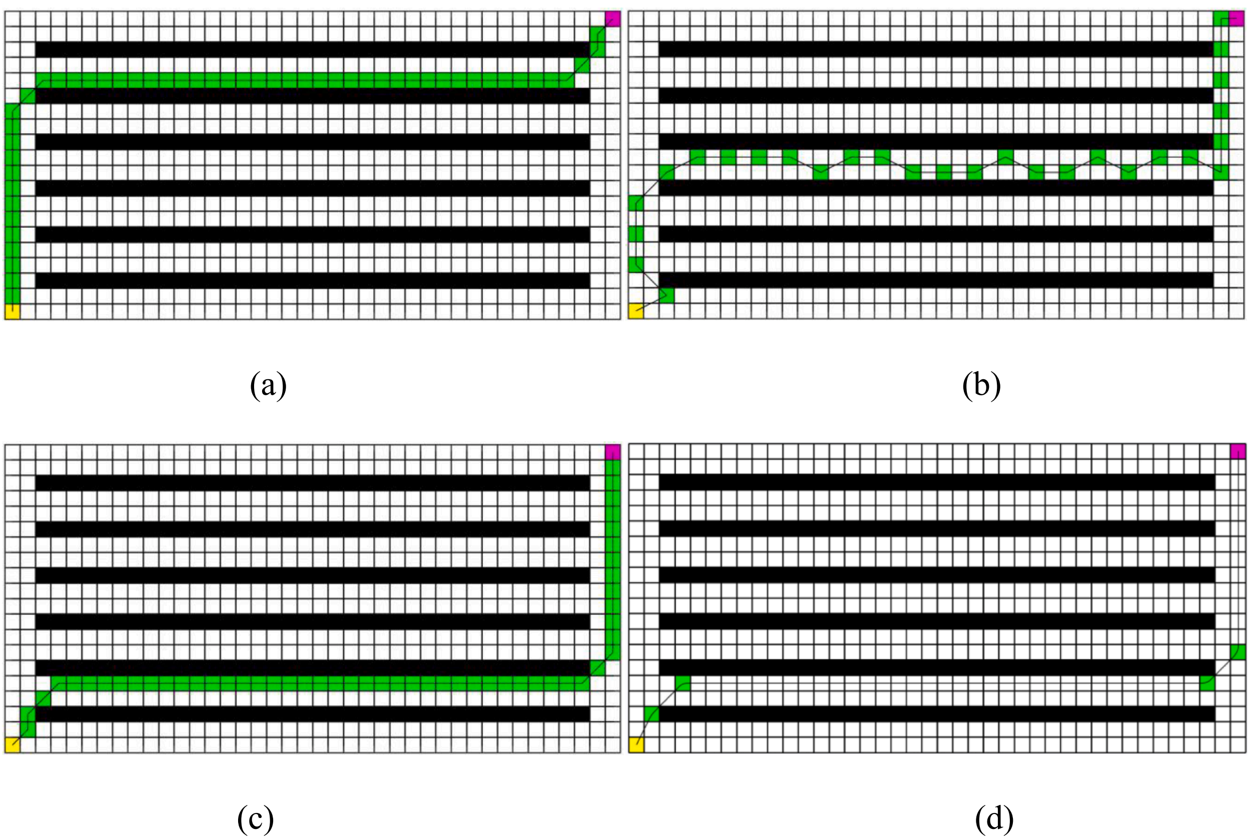


Fig. 12. The path planning simulation results of different algorithms in theoretical greenhouse: (a) Dijkstra algorithm, (b) RRT algorithm, (c) A* algorithm, (d) Improved A* algorithm.

$$\begin{cases} x_{t+1} = x_t + v\Delta t\cos\theta_t \\ y_{t+1} = x_t + v\Delta t\sin\theta_t \\ \theta_{t+1} = \theta_t + \omega\Delta t \end{cases} \tag{8}$$

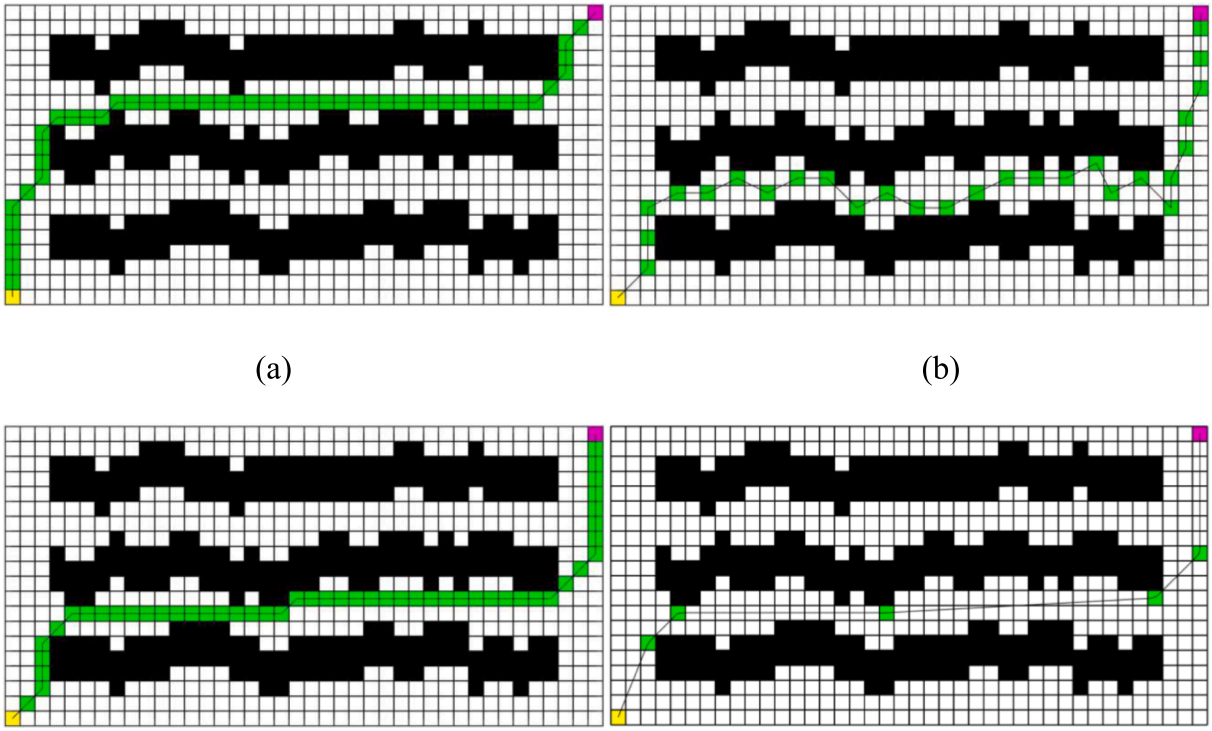
Where: x_t, y_t, x_{t+1}, y_{t+1} are the x and y coordinate positions of the tractor at time t and $t+1$ respectively; θ_t and θ_{t+1} are the heading angles of the tractor at time t and $t+1$, respectively. v is the linear speed of the tractor; ω is the angular speed of the driverless car; Δt is the time interval from time t to time $t+1$.

3.2. Speed sampling

The DWA algorithm limits the speed selection problem of the tractor when moving to the speed vector space composed of the linear speed and angular velocity of the tractor, and it considers the dynamic constraints and nonholonomic constraints of the tractor in the speed vector space. Meanwhile, the algorithm samples the speed vector space of the tractor and then evaluates the quality of the generated trajectory to determine the optimal trajectory. Therefore, the position control problem of the tractor is transformed into the speed control problem of the tractor, and the motion of the tractor is controlled by outputting the real-time optimal speed. In actual working environments, the tractor's speed is affected by the maximum and minimum speed of the tractor, the performance of the motor, and the braking distance. The above three conditions can constrain the speed of the tractor within a certain range. The specific constraints are given below:

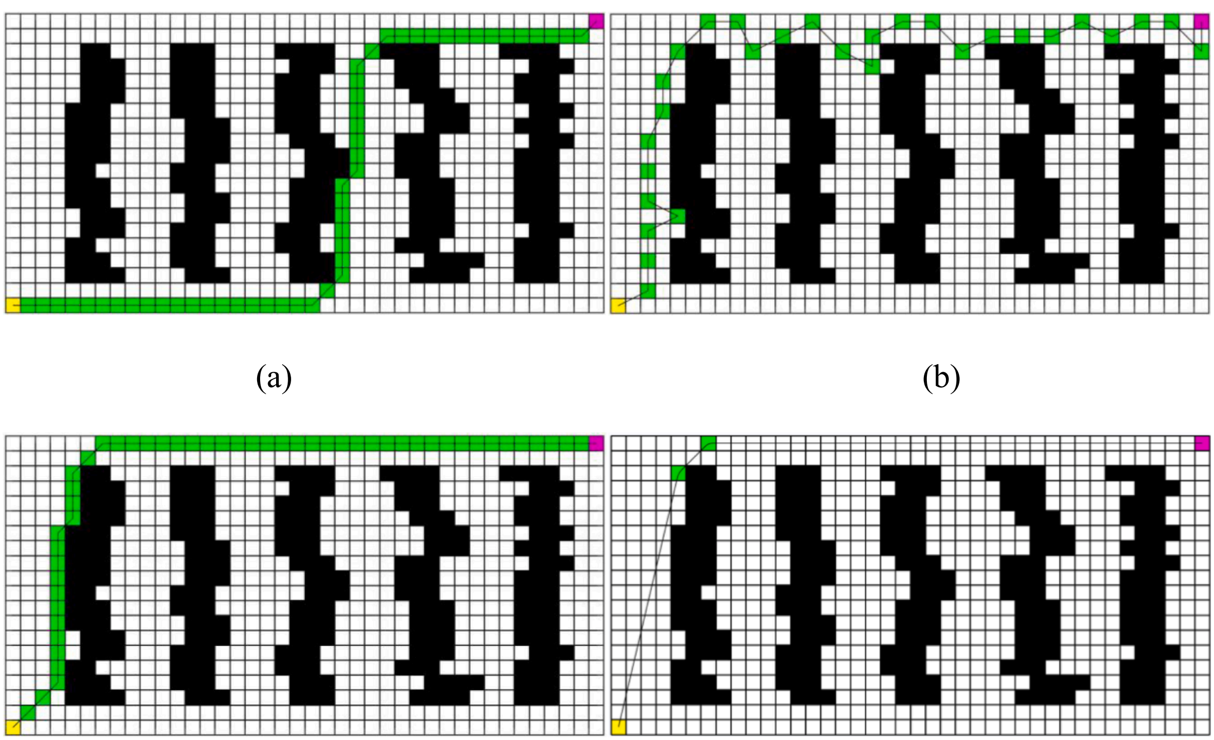
(1) Maximum and minimum speed constraints

Since the hardware performance of the tractor is limited by the minimum and maximum speed of the tractor, the sampling speed of the



(a) (b) (c) (d)

Fig. 13. The path planning simulation results of different algorithms in north–south ridge greenhouse: (a) Dijkstra algorithm, (b) RRT algorithm, (c) A* algorithm, (d) Improved A* algorithm.



(a) (b) (c) (d)

Fig. 14. The path planning simulation results of different algorithms in east–west ridge greenhouse: (a) Dijkstra algorithm, (b) RRT algorithm, (c) A* algorithm, (d) Improved A* algorithm.

Table 3
Comparison of the runtime, turn points, and path length of different path planning algorithms in simulated greenhouse scene.

Parameter	Greenhouse environment	Dijkstra algorithm	RRT algorithm	Traditional A* algorithm	Improved A* algorithm	RR1/%	RR2/%	RR3/%
Runtime/s	The theoretical	0.070571	0.091395	0.059024	0.056246	20.30	38.46	4.71
	The north–south ridge	0.084244	0.094500	0.059234	0.056475	32.96	40.24	4.66
	The east–west ridge	0.080054	0.099288	0.058170	0.055703	30.42	43.90	4.24
	average	/	/	/	/	27.89	40.86	4.54
Number of turn points	The theoretical	5	20	5	4	20.00	80.00	20.00
	The north–south ridge	9	22	7	5	44.44	77.27	28.57
	The east–west ridge	7	25	5	2	71.43	92.00	60.00
	average	/	/	/	/	45.29	83.09	36.19
Path length/cm	The theoretical	59.1803	61.2536	55.0711	53.4234	9.73	12.78	2.99
	The north–south ridge	53.3137	60.5542	53.3137	52.4840	1.56	13.33	1.56
	The east–west ridge	54.4853	66.6187	54.4853	53.2927	2.19	20.00	2.19
	average	/	/	/	/	4.49	15.37	2.25

Note: RR_1 is the reduction rate of improved A* compared with Dijkstra algorithm, RR_2 is the reduction rate of improved A* compared with RRT algorithm, and RR_3 is the reduction rate of improved A* compared with traditional A* algorithm.

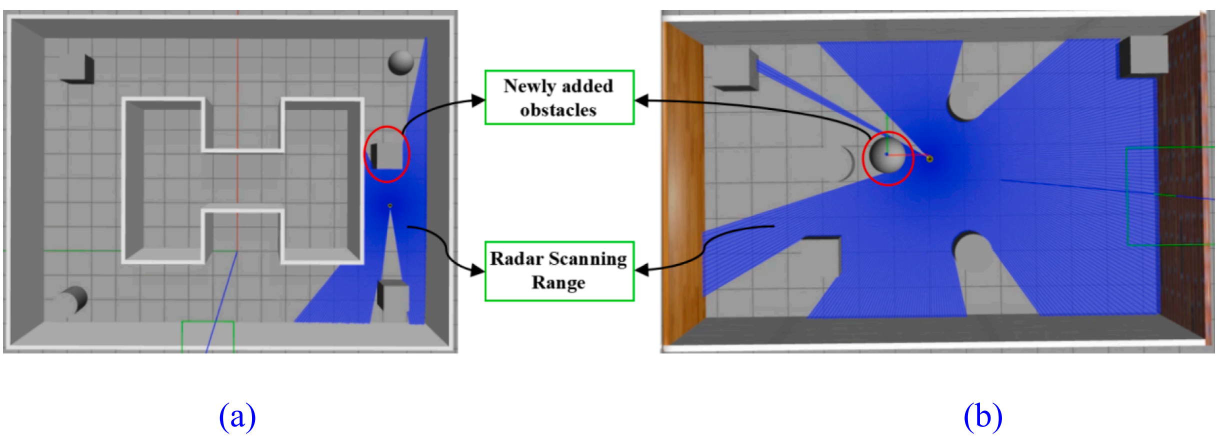


Fig. 15. Adding unknown obstacles in the simulation environment: (a) Adding unknown obstacles to the "I-zigzag" map, (b) Adding unknown obstacles to the "rectangular" map.

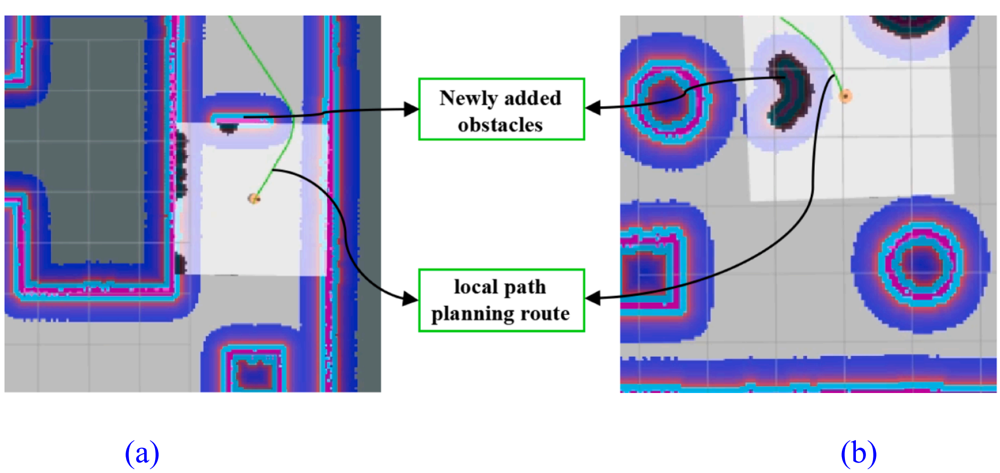


Fig. 16. Local route planning route: (a) Dynamic obstacle avoidance in the "I-zigzag" map, (b) Dynamic obstacle avoidance in the "rectangular" map.

tractor should be controlled within the optimal angular velocity and linear velocity range of the tractor. The constraint is:

$$V_m = \{(v,\omega)|v \in [v_{\min}, v_{\max}], \omega \in [\omega_{\min}, \omega_{\max}]\} \quad (9)$$

Where, $v_{\max}$ is maximum linear velocity of the tractor; $v_{\min}$ is the minimum linear velocity; $\omega_{\max}$ is maximum angular velocity of the tractor; $\omega_{\min}$ is the minimum angular velocity.

(2) Motor acceleration and deceleration constraints

The acceleration of the tractor is subject to the output torque of the motor. The velocity vector space sampling should be conducted within the tolerable range of the motor torque. The constraint is:

$$V_d = \left\{(v,\omega)\middle| v \in \left[v_c - \dot{v}_b\Delta t, v_c + \dot{v}_a\Delta t\right], \omega \in \left[\omega_c - \dot{\omega}_b\Delta t, \omega_c + \dot{\omega}_a\Delta t\right]\right\} \quad (10)$$

where, v_c is current linear velocity of the tractor; ω_c is current

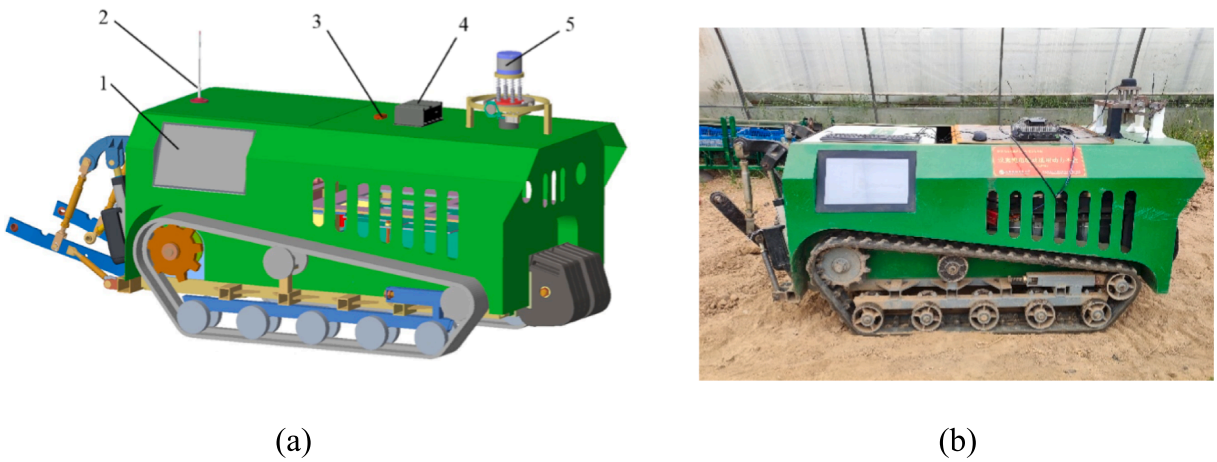


Fig. 17. The overall composition of the autonomous navigation electric tractor: (a) 3D schematic, (b) Actual vehicle diagram. 1.On-board Display; 2. Antenna; 3. IMU; 4. Industrial Control Unit; 5.LIDAR.

Fig. 18. The overall drawing of the facility greenhouse.

angular velocity of the tractor; $\dot{v}_a$ is maximum linear acceleration of the tractor; $\dot{\omega}_a$ is maximum angular acceleration of the tractor; $\dot{v}_b$ is maximum line deceleration of the tractor; $\dot{\omega}_b$ is maximum angular deceleration of the tractor.

(3) Braking distance constraint

When the tractor encounters a dynamic obstacle in the local path planning, the safe distance between the tractor and the obstacle should be guaranteed. To ensure that the tractor stops before hitting a random obstacle, under the maximum deceleration condition, the linear velocity and angular velocity of the tractor are reduced to 0 before the tractor reaches the obstacle. The constraint is:

$$V_a = \left\{ (v,\omega), v \le \sqrt{2\mathrm{dist}(v,\omega)\dot{v}_b}, \omega \le \sqrt{2\mathrm{dist}(v,w)\dot{\omega}_b} \right\} \tag{11}$$

Where, dist(v, ω) is the nearest distance of tractor to obstacle.

The final tractor speed is subjected to the intersection of the above three constraints, i.e., the dynamic window speed V_r satisfies Eq. (12):

$$V_r = V_m \cap V_d \cap V_a \tag{12}$$

3.3. Evaluation function

In the sampled velocity vector space, by sampling the continuous velocity vector space V_r matching the kinematic model and discretizing the discrete sampling points, there are many groups of feasible trajectories excluding the trajectories colliding with the obstacles. Thus, it is necessary to evaluate the simulated trajectories in the set of velocity groups and screen out the trajectories with the best performance to be sent to the tractor's chassis, and then control the tractor to complete the evasion task. task. The trajectory is evaluated by a traditional evaluation function as follows:

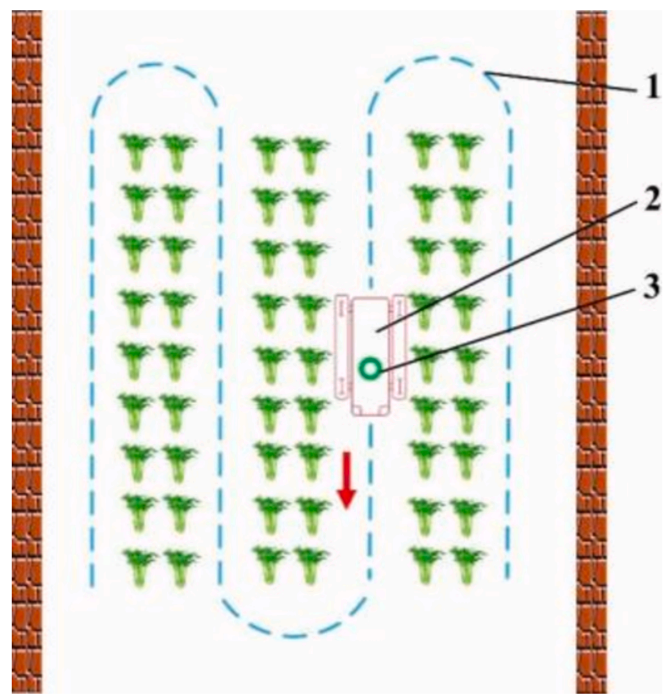


Fig. 19. The schematic diagram of the navigation path.

$$G(v,\omega) = \sigma(\alpha \cdot heading(v,\omega) + \beta \cdot dist(v,\omega) + \gamma \cdot velocity(v,\omega)) \tag{13}$$

The evaluation function of the DWA algorithm involves deflection angle, linear velocity, and the closest distance between the end of the simulated trajectory and the obstacle. Since these three indicators have different measurement units, to avoid that a certain evaluation function accounts for a too large proportion and leads to an unsmooth trajectory, it is necessary to normalize each evaluation function and then add them

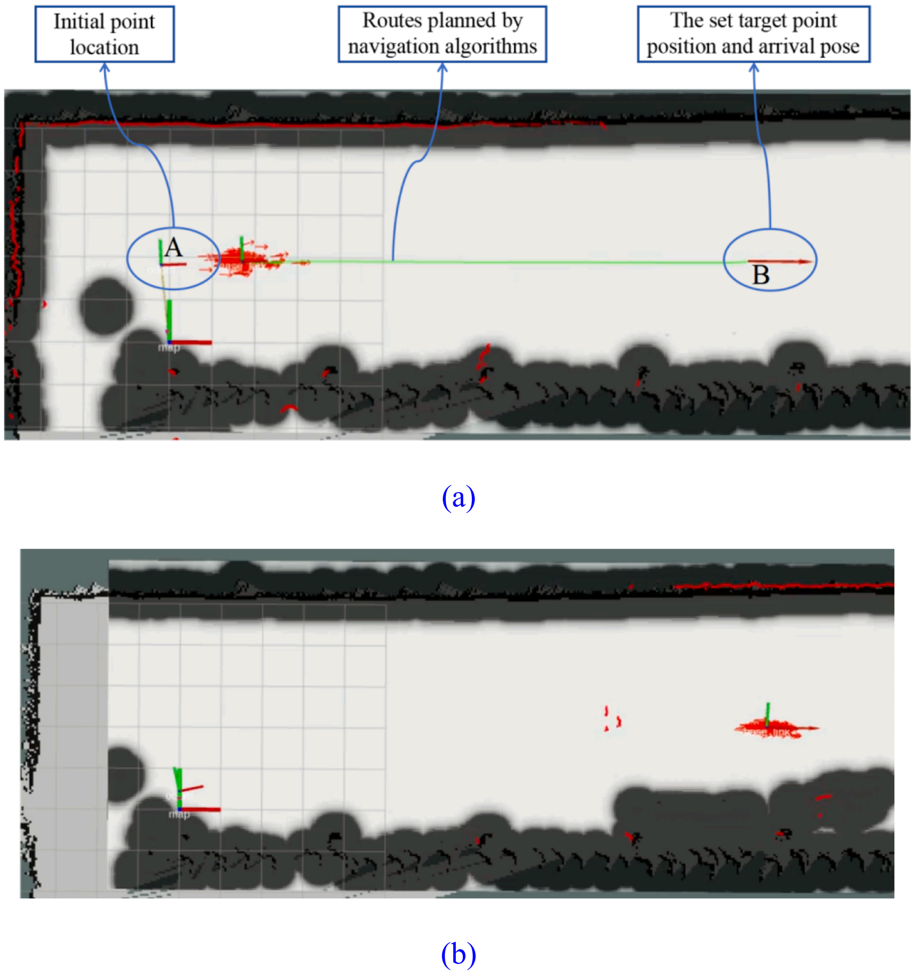


Fig. 20. Navigation test: (a) Route of path planning in Rviz, (b) Display of arrival at the target position in Rviz.

Table 4
Navigation accuracy test results.

Test number	Lateral deviation /cm			Course deviation /°		
	Maximum value	Mean value	Standard deviation	Maximum value	Mean value	Standard deviation
1	+11.20	+6.44	4.87	−11	−4.37	5.53
2	+10.80	+4.99	5.75	−13	−5.64	6.82
3	−9.80	−4.95	5.62	−12.60	−4.90	6.16
4	−10.50	−5.35	6.20	−11.60	−5.15	6.16
5	+9.60	+5.67	4.47	10.60	+5.66	6.57
Mean value	+10.38	+5.48	5.382	−11.76	−5.144	6.248

together. The normalization is to divide each item by the sum of all items:

$$normal\_heading(i) = \frac{heading(i)}{\sum_{i=1}^{n} heading(i)} \tag{14}$$

$$normal\_dist(i) = \frac{dist(i)}{\sum_{i=1}^{n} dist(i)} \tag{15}$$

$$normal\_velocity(i) = \frac{velocity(i)}{\sum_{i=1}^{n} velocity(i)} \tag{16}$$

where, heading(v,ω) is the azimuthal evaluation subfunction, which evaluates the azimuthal deviation between the direction of the end of the simulated trajectory and the target at the current velocity As shown in the figure, heading(v,ω) = π-θ; dist(v,ω) is obstacle evaluation subfunction, representing the closest distance of the simulated trajectory of the corresponding velocity group to an obstacle, and dist(v,ω) is set to a constant if there is no obstacle on the current trajectory; velocity(v,ω) is the velocity evaluation subfunction, representing the magnitude of the velocity of the simulated trajectory; σ is the smooth function; α is the direction influence coefficient, the value of which is related to the forward direction of the tractor, as the tractor gets closer to the target point, the value of α increases accordingly;β is the safety distance coefficient, the value of which is related to the distance of the tractor from the obstacle, as the distance of the tractor from the obstacle decreases, the value of β increases accordingly. Γ is the speed influence coefficient, the value of which is related to the speed of the tractor, the greater the speed of the tractor, the greater will be the value of γ (see Fig. 6).

3.4. Improvement of evaluation function

The evaluation function of the DWA algorithm is crucial for assessing

Fig. 21. Arrangement of obstacles.

the viability of simulated trajectories and for selecting the optimal pathway. However, the traditional DWA algorithm presents significant limitations; it fails to incorporate global path information during the obstacle avoidance process, which often results in convergence to local optima. To enhance the algorithm's effectiveness, this paper suggests modifying the evaluation function to integrate relevant information from the optimized global optimal path. By doing so, the tractor will be able to adhere more closely to the globally optimal path while effectively navigating around obstacles. The proposed improvements to the DWA evaluation function are outlined as follows:

$$G(\nu,\omega) = \sigma(\alpha \cdot key(\nu,\omega) + \beta \cdot dist(\nu,\omega) + \gamma \cdot velocity(\nu,\omega)) \tag{17}$$

Where: $dist_1(\nu,\omega)$ denotes the Euclidean distance from the end of the simulated trajectory to the local target, and η denotes the weight of the newly added evaluation index.

The enhanced dynamic window approach is designed to closely align with the planned global optimal path while ensuring effective obstacle avoidance. Fig. 7 in the document illustrates the flowchart of the improved algorithm.

4. Algorithm fusion

The improved A* algorithm can obtain the optimal solution in global path planning. It performs well in global path planning in a static working environment but fails to realize local real-time obstacle avoidance when encountering unknown obstacles. The DWA algorithm can solve the local obstacle avoidance problem well, but it is easy to fall into the local optimum when there are more obstacles in the environment due to the guidance of only one final goal point, which will result in a larger global path or even a path planning failure. To overcome the defects of the two optimization algorithms, this paper fuses the improved A* algorithm and the improved DWA algorithm, extracts the key point of the global path planned by the improved A* algorithm as the intermediate goal point of the DWA algorithm to guide the local path planning; the optimized evaluation function makes the local path planning follow the planned global path contour to avoid the excessive deviation of local planning. The fusion algorithm guarantees that the global path is optimal while ensuring good obstacle avoidance and movement effects during local planning. The flowchart of the fusion algorithm is shown in Fig. 8.

The sequence of operations for the tractor fusion path planning algorithm presented in this paper is structured as follows:

1) The tractor gathers information about the operating environment and constructs a global map.

2) The tractor establishes its initial position and the target location, then employs the enhanced A* algorithm described in this paper to conduct global path planning. The algorithm stores the coordinates of each key point optimized during this planning phase, designating each as a temporary target point.

3) The tractor proceeds along the planned path, continuously assessing the presence of unknown obstacles. If no obstacles are detected, it continues moving; if unknown obstacles are encountered, the improved DWA algorithm is utilized for local path planning.

4) As the tractor progresses, it checks whether it has surpassed a temporary target point. Upon passing, it updates the temporary target point to the next node, which serves as the new temporary target for subsequent local path planning.

5) The process concludes by checking if the final target position has been reached. If the tractor arrives at the final target point, it stops; if not, it continues moving along the planned path.

5. Algorithm simulation and test

5.1. Simulation test in different environments

In order to verify the effectiveness of this algorithm, three different grid map scenes were constructed with the same specifications, and the complexity and proportion of obstacles gradually increased. Dijkstra algorithm, RRT algorithm, A* algorithm and improved A* algorithm were compared in three different grid maps, and the same starting and ending coordinates were set. The yellow grid represents the start point, the purple grid represents the end point, the white grid is the movable space, the black grid is the obstacle, the green grid is the path node, and the black solid line represents the generated global path. The running environment of all simulation experiments is 64-bit windows 11, the PC is Lenovo Rescuer R9000P, the processor is R7-5800H, the memory is 16G, and the Matlab version is 2016a (see Figs. 9–11).

To ensure the accuracy of the results, for these three different maps,

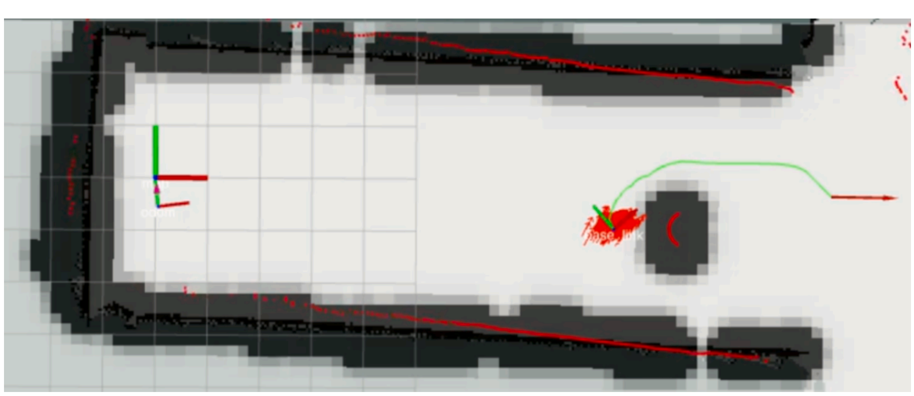


Fig. 22. Obstacle avoidance process in the field test.

four types of path planning algorithms are used, each of which runs 10 times, and the initial setup parameters are the same. The evaluation indexes are the running time, the number of inflection points, and the length of the path. Each index is counted in each of the 10 groups for each algorithm and each map, and the average value is taken, as shown in Table 2.

It can be seen from Table 2 that the runtime and path length planned by the improved A* algorithm are shorter and the number of turn points is less. Compared with Dijkstra algorithm, RRT algorithm and traditional A* algorithm, the average runtime of the improved A* algorithm is respectively reduced by 18.49 %, 30.08 % and 7.66 %. The path length is respectively shortened by 1.57 %, 17.86 % and 3.51 %. The number of inflection points is respectively decreased by 31.58 %, 51.52 % and 44.49 %. The improved A* algorithm can obtain the global path optimization, and the global path planned by the improved A* algorithm is smoother, the path length and runtime are shorter, and it is convenient for the tractor to execute the path trajectory to the target point.

In order to test the advantages of the improved A* algorithm in the greenhouse environment, three simulated greenhouse grid maps with the same specifications were set up, namely, the simulated theoretical greenhouse, the north–south ridge greenhouse and the east–west ridge greenhouse close to the real environment. The simulation experiment results are shown in Figs. 12-14 and Table 3.

It can be seen from Table 3, in the three simulated greenhouse scenarios, the path planned by the improved A* algorithm is better. Compared with Dijkstra algorithm, RRT algorithm and traditional A* algorithm, the average runtime is respectively shortened by 27.89 %, 40.86 % and 4.54 %. The path length is respectively shortened by 4.49 %, 15.37 % and 2.25 %. The number of inflection points is respectively decreased by 45.29 %, 83.09 % and 36.19 %. Therefore, the improved A* algorithm is used in this study for global path planning of electric tractors in greenhouse.

5.2. *Analysis of stochastic obstacle avoidance by fusion of improved A* algorithm and DWA algorithm*

Install Navigation function package in ROS, add and modify related nodes. In Navigation, move_base loads map and path planning plug-ins through preset interfaces, reads setting parameters, and publishes related speed command topics. The /base_control node accepts speed instructions issued by move_base. In the Gazebo simulation environment, this study builds “I-zigzag” and “rectangular” maps and adds temporary unknown obstacles to verify the effect of local path planning. In the navigation process, when the tractor runs smoothly along the planned path, temporary obstacle cartons are added to the planned route to simulate temporary obstacles, as illustrated in Fig. 15.

The local path planning route is shown in Fig. 16. When a temporary obstacle appears, the LiDAR sensor scans for information about the obstacle and passes the laser point cloud information into the local path planning algorithm. Then, the algorithm reprograms the tractor’s chassis speed and selects a new navigation path.

The simulation test of the fusion algorithm shows that the tractor completes the obstacle avoidance task smoothly by the dynamic speed group planned in real-time, it can pass at the obstacle bypass point smoothly and reach the set target point accurately, and the planned path can keep the global optimum. These results demonstrate the feasibility of the fusion algorithm and show that it can be applied to the electric tractor.

5.3. *Experiments on electric tractor navigation in facility greenhouses*

5.3.1. *Test condition*

In this study, a self-developed small electric crawler tractor is used as the power chassis, employing differential steering mode for maneuverability. The DTB-3049-H310 industrial computer serves as the upper computer, equipped with an i3-9100 T processor, 8 GB of RAM, and a 128 GB hard drive, running Ubuntu 16.04 and ROS Kinetic. The navigation system incorporates a two-dimensional Lidar and a 9-axis inertial measurement unit to gather environmental information and determine the chassis pose. The Lidar, a RPLIDAR A2M7 model from Silan Technology, operates at a scanning frequency of 10 Hz and has a measurable range of 0.2 m to 16 m. The IMU, a HIFI-A9 model from TAObotics Company, achieves a return frequency of 300 Hz. Additionally, a 15-inch onboard display is installed on the chassis to facilitate dynamic monitoring of the tractor and provide navigation guidance (see Fig. 17).

The test site is situated in the demonstration greenhouse of the Kneogu Fruit and Vegetable Professional Cooperative, located in the Yangling District, Shaanxi Province, China, as depicted in Fig. 18. The facility’s environment is relatively enclosed, featuring a centrally located cement road that is evenly distributed and relatively flat. Vegetable planting areas flank both sides of this road.

When the electric crawler tractor is operating in the greenhouse, it needs to drive straight along the area to be operated, drive to the ground to stop the operation, turn the ground, and handle the next operation area. The job navigation path is illustrated in Fig. 19.

5.3.2. *Navigation test*

(1) Straight-line navigation test.

The navigation process is shown in Fig. 20, the improved A* algorithm plans an optimal path from the starting point to the target point. The current position is the starting point of the navigation, the red arrow denotes the set target point position and the arrival pose orientation, the red line represents the laser radar scanning point, the black line represents the expanded obstacle information, and the green line represents the planned global path. After setting the target point and pointing, the tractor begins to automatically move from the starting point along the planned path to the target point and adjust the posture, as illustrated in Fig. 20(b). The start point of path planning is A, and the target point is B.

It can be seen from Fig. 20 that the route planned by the improved A* algorithm is smooth; meanwhile, the tractor can navigate autonomously according to the path planned by the navigation system, and it can reach the target point at a stable speed and complete the specified pose accurately, which verifies the effectiveness of the navigation algorithm proposed in this study.

To test the accuracy and stability of the tractor navigation system, the straight path tracking test is conducted in the process of tractor navigation. The speed of the tractor is 1 m/s. Five groups are set in the test, and the test distance is 20 m. In each group, data sampling measurement is performed at an interval of 2 m. After the navigation, the lateral deviation and heading deviation of the tractor at each sampling point are counted. It is stipulated that the tractor reaches the left side of the target point as a positive deviation and reaches the right side of the target point as a negative deviation. The actual heading is a positive deviation on the left side of the target heading and a negative deviation on the right side of the target heading. The average deviation and standard deviation of the tracking path are calculated according to the test data. After each test, the initial position of the tractor is reset to prevent the cumulative error from affecting the test results. The test results are listed in Table 4.

The test results indicate that the maximum lateral deviation is 11.20 cm, the maximum average lateral deviation is 6.44 cm, and the average standard deviation is 5.382 cm after five tests. Meanwhile, the maximum heading deviation is 13°, the maximum average heading deviation is 5.66°, and the average standard deviation is 6.248°, which can meet the accuracy requirements of the tractor’s autonomous navigation in the facility greenhouse.

(2) Obstacle avoidance test.

For the occasional temporary obstacles, based on the global path planning of the tractor, the local moving path is dynamically adjusted according to the obstacle information obtained by the lidar. To verify the effectiveness of the fusion algorithm, a field experiment of obstacle avoidance is carried out in a facility greenhouse, and the result is shown in Fig. 21.

After the lidar detects the obstacle, it will mark the obstacle's position on the grid map. Once obstacles appear on the global path, local path planning will plan a new path around the obstacles. Once the obstacle is avoided, the tractor will return to the global path until it finally reaches the target point. The real-time obstacle avoidance process of the tractor is illustrated in Fig. 22.

The above obstacle avoidance experiments show that the fusion algorithm proposed in this study can solve the obstacle avoidance problem of tractors in facility greenhouses, which proves its feasibility. Through local path planning, the tractor can avoid accidental obstacles when navigating in the facility greenhouse, which brings great convenience to the use of the tractor in the facility greenhouse. These results indicate that the application of the proposed method can improve the tractor's adaptability to the facility environment and the feasibility of improving the intelligence of agricultural machinery.

6. Conclusions

To improve the intelligent level of electric tractors in the facility greenhouse environment, this paper proposes a path planning method combining the improved A* algorithm and DWA algorithm and applies it to the autonomous navigation of electric crawler tractors in the greenhouse. Simulation experiments demonstrate that compared with the traditional A* algorithm, the improved A* algorithm can not only obtain the global optimal path but also obtain a smoother planned global path. In common scenarios, the average runtime is shortened by 7.66 %, the average path length is reduced by 3.51 %, and the average number of turn points is reduced by 44.49 %. In the simulation greenhouse scenarios, the average runtime is shortened by 4.54 %, the average path length is reduced by 2.25 %, and the average number of turn points is reduced by 36.19 %. The fusion algorithm has good obstacle avoidance effect. A real vehicle path planning experiment was conducted in the facility environment. The results indicate that the route planned by the improved A* algorithm is smoother. The maximum lateral deviation is 11.20 cm, and the maximum heading deviation is 13°. Meanwhile, the fusion algorithm exhibits a good obstacle avoidance effect. It can complete the obstacle avoidance task while obtaining the global optimal path, which can meet the accuracy requirement of the autonomous navigation of the electric tractor in the facility greenhouse environment.

CRediT authorship contribution statement

Huiping Guo: Writing – review & editing, Supervision, Project administration, Investigation. **Yi Li:** Writing – original draft, Investigation, Data curation. **Hao Wang:** Writing – original draft. **Chensi Wang:** Resources, Data curation. **Jiao Zhang:** Methodology, Data curation. **Tingwei Wang:** Validation, Data curation. **Linrui Rong:** Validation, Data curation. **Haoyu Wang:** Writing – original draft. **Zihao Wang:** Writing – original draft. **Yaobin Huo:** Visualization. **Shaomeng Guo:** Visualization. **Fuzeng Yang:** Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Key R&D Program of China (No. 2023YFD2000300), the Shaanxi Provincial Technology Innovation Guidance Special Project (No. 2021QFY08-05), Zhejiang province agricultural machinery Research, manufacturing and application integration project (No. YF20220801) and the Shaanxi Province Key Research and Development Plan Project (No. 2022ZDLNY03-04).

Data availability

Data will be made available on request.

References

Cabreira, T.M., Brisolara, L.B., Ferreira Jr., P.R., 2019. Survey on coverage path planning with unmanned aerial vehicles. Drones 3 (1), 4. https://doi.org/10.3390/drones3010004.

Cao, R., Zhang, Z., Li, S., Zhang, M., Li, H., Li, M., 2021. Multi machine collaborative global path planning based on improved A* algorithm and Bezier curve. Transactions of the Chinese Society of Agricultural. Machinery 52, 548–554 in Chinese.

Cao, X., Zou, X., Jia, C., Chen, M., Zeng, Z., 2019. RRT-based path planning for an intelligent litchi-picking manipulator. Comput. Electron. Agric. 156, 105–118. https://doi.org/10.1016/j.compag.2018.10.031.

Cheng, C., Hao, X., Li, J., Zhang, Z., Sun, G., 2017. Global Dynamic Path Planning Based on Fusion of Improved A* Algorithm and Dynamic Window Approach. J. Xi'an Jiaotong Univ. 51 (11), 137–143 in Chinese.

Chu, K., Lee, M., Sunwoo, M., 2012. Local Path Planning for Off-Road Autonomous Driving With Avoidance of Static Obstacles. IEEE Trans. Intell. Transp. Syst. 13 (4), 1599–1616. https://doi.org/10.1109/TITS.2012.2198214.

Duan, L., Yang, F., Yan, B., Shi, S., Qin, J., 2022. Research progress of apple production intelligent chassis and weeding and harvesting equipment technology. Smart Agric 4 (03), 24–41 in Chinese.

Fox, D., Burgard, W., Thrun, S., 1997. The dynamic window approach to collision avoidance. IEEE Rob. Autom. Mag. 4 (1), 23–33. https://doi.org/10.1109/100.580977.

González, D., Pérez, J., Milanés, V., Nashashibi, F., 2016. A Review of Motion Planning Techniques for Automated Vehicles. IEEE Trans. Intell. Transp. Syst. 17 (4), 1135–1145. https://doi.org/10.1109/TITS.2015.2498841.

Huang, C., Fei, J., Liu, Y., Li, H., Liu, X., 2017. Smooth path planning method based on dynamic feedback A* ant colony algorithm. Nongye Jixie Xuebao/transactions of the Chinese Society of Agricultural Machinery 48 (4), 34–40 in Chinese.

Jeon, C.-W., Kim, H.-J., Yun, C., Gang, M., Han, X., 2021. An entry-exit path planner for an autonomous tractor in a paddy field. Comput. Electron. Agric. 191, 106548. https://doi.org/10.1016/j.compag.2021.106548.

Jin, M., Wang, H., 2023. Robot path planning by integrating improved A* algorithm and DWA algorithm. J. Phys. Conf. Ser. 2492 (1), 012017. https://doi.org/10.1088/1742-6596/2492/1/012017.

Lao, C., Li, P., Feng, Y., 2021. Path planning of greenhouse robot based on fusion of improved A* algorithm and dynamic window approach. Nongye Jixie Xuebao/transactions of the Chinese Society of Agricultural Machinery 52 (1), 14–22 in Chinese.

Li, B., Zhang, Y., Ouyang, Y., Liu, Y., Zhong, X., Cen, H., Kong, Q., 2021. Fast Trajectory Planning for AGV in the Presence of Moving Obstacles: A Combination of 3-dim A* Search and QCQP. In: In: 2021 33rd Chinese Control and Decision Conference (CCDC). Presented at the 2021 33rd Chinese Control and Decision Conference (CCDC), pp. 7549–7554. https://doi.org/10.1109/CCDC52312.2021.9602686.

Liu, L., Wang, B., Xu, H., 2022. Research on Path-Planning Algorithm Integrating Optimization A-Star Algorithm and Artificial Potential Field Method. Electronics 11 (22), 3660. https://doi.org/10.3390/electronics11223660.

Liu, L., Wang, X., Yang, X., Liu, H., Li, J., Wang, P., 2023. Path planning techniques for mobile robots: Review and prospect. Expert Syst. Appl. 227, 120254. https://doi.org/10.1016/j.eswa.2023.120254.

Liu, J., Xue, L., Zhang, H., Liu, Z., 2021. Robot Dynamic Path Planning Based on Improved A* and DWA Algorithm. Comput. Eng. Appl. 57 (15), 73–81 in Chinese.

Mao, W., Liu, H., Wang, X., Yang, F., Liu, Z., Wang, Z., 2022. Design and Experiment of Dual Navigation Mode Orchard Transport Robot. Nongye Jixie Xuebao/transactions of the Chinese Society of Agricultural Machinery 53 (03), 27–39+49 in Chinese.

Meng, B.H., Godage, I.S., Kanj, I., 2022. RRT*-Based Path Planning for Continuum Arms. IEEE Rob. Autom. Lett. 7 (3), 6830–6837. https://doi.org/10.1109/LRA.2022.3174257.

Nasiboglu, R., 2022. Dijkstra solution algorithm considering fuzzy accessibility degree for patch optimization problem. Appl. Soft Comput. 130, 109674. https://doi.org/10.1016/j.asoc.2022.109674.

Ou, Y., Fan, Y., Zhang, X., Lin, Y., Yang, W., 2022. Improved A* Path Planning Method Based on the Grid Map. Sensors 22 (16), 6198. https://doi.org/10.3390/s22166198.

Tan, C., Li, Y., Mao, W., Yang, F., 2020. Research Progress on Automatic Navigation Technology of Agricultural Machinery. Journal of Agricultural Mechanization. Research 42 (05), 7–14+32 in Chinese.

Tang, G., Tang, C., Claramunt, C., Hu, X., Zhou, P., 2021. Geometric A-Star Algorithm: An Improved A-Star Algorithm for AGV Path Planning in a Port Environment. IEEE Access 9, 59196–59210. https://doi.org/10.1109/ACCESS.2021.3070054.

Wang, Y., Liu, D., Zhao, H., Li, Y., Song, W., Liu, M., Tian, L., Yan, X., 2022. Rapid citrus harvesting motion planning with pre-harvesting point and quad-tree. Comput. Electron. Agric. 202, 107348. https://doi.org/10.1016/j.compag.2022.107348.

Wang, P., Liu, Y., Yao, W., Yu, Y., 2023. Improved A-star algorithm based on multivariate fusion heuristic function for autonomous driving path planning. Proc. Institut. Mech. Eng., Part D: J. Automobile Eng. 237 (7), 1527–1542. https://doi.org/10.1177/09544070221100677.

Wu, B., Chi, X., Zhao, C., Zhang, W., Lu, Y., Jiang, D., 2022. Dynamic Path Planning for Forklift AGV Based on Smoothing A* and Improved DWA Hybrid Algorithm. Sensors 22 (18), 7079. https://doi.org/10.3390/s22187079.

Wu, F., Guo, S., 2020. Dynamic Path Planning of AGV Based on Fusion of Improved A* and Dynamic Window Approach. Science Technology and Engineering 20 (30), 12452–12459 in Chinese.

Xin, Y., Liang, H., Du, M., Mei, T., Wang, Z., Jiang, R.H., 2014. An improved A* algorithm for searching infinite neighbourhoods. Robot 36, 627–633 in Chinese.

Xiong, X., Min, H., Yu, Y., Wang, P., Xiong, X., Min, H., Yu, Y., Wang, P., 2021. Application improvement of A* algorithm in intelligent vehicle trajectory planning. Math. Biosci. Eng. 18 (1), 1–21. https://doi.org/10.3934/mbe.2021001.

Yang, F., Liu, J., 2021. Path Planning of Mobile Robot Combining improved A* Algorithm and Dynamic Window Algorithm. Industrial Conirol Computer 34 (5), 106–112 in Chinese.

Yi, J., Yuan, Q., Sun, R., Bai, H., 2022. Path planning of a manipulator based on an improved P_RRT* algorithm. Complex Intell. Syst. 8 (3), 2227–2245. https://doi.org/10.1007/s40747-021-00628-y.

Yin, X., Cai, P., Zhao, K., Zhang, Y., Zhou, Q., Yao, D., 2023. Dynamic Path Planning of AGV Based on Kinematical Constraint A* Algorithm and Following DWA Fusion Algorithms. Sensors 23 (8), 4102. https://doi.org/10.3390/s23084102.

Zhang, W., Liu, Y., Zhang, C., Zhang, L., Xia, Y., 2017. Real-time Path Planning of Greenhouse Robot Based on Directional A* Algorithm. Nongye Jixie Xuebao/ Transactions of the Chinese Society of Agricultural Machinery 2017,48(7),22-28. (in Chinese).

Zhang, Z., Liu, X., 2015. The present situation and countermeasures of facility agriculture development in our country. Issues Agric. Econ. 36 (05), 64–70+111 in Chinese.

Zhao, X., Wang, Z., Huang, C., 2018. Mobile robot path planning based on improved A * algorithm. Robot 40 (6), 903–910 in Chinese.

Zhao, X., Wang, K., Wu, S., Wen, L., Chen, Z., Dong, L., Sun, M., Wu, C., 2023. An obstacle avoidance path planner for an autonomous tractor using the minimum snap algorithm. Comput. Electron. Agric. 207, 107738. https://doi.org/10.1016/j.compag.2023.107738.

Zhou, J., He, Y., 2021. Research progress on navigation path planning of agricultural machinery. Nongye Jixie Xuebao/transactions of the Chinese Society of Agricultural Machinery 52 (09), 1–14 in Chinese.